

Lattice — Feature Showcase

Open this file with **Lattice** to see every feature rendered live.

Switch between **Edit**, **Preview**, and **Dual** view with the toolbar selector.

Try `Ctrl/Cmd+Shift+T` to auto-generate the Table of Contents below.

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Text Formatting

Normal paragraph text flows here. Lattice renders Markdown as **close to GitHub** as possible, so files look identical whether viewed in the editor preview, on GitHub, or printed.

Style	Syntax	Result
Bold	<code>**bold**</code>	bold
Italic	<code>*italic*</code>	<i>italic</i>
Bold + italic	<code>***both***</code>	<i>both</i>
Strikethrough	<code>~~strikethrough~~</code>	strikethrough
Inline code	<code>`code`</code>	<code>code</code>
Superscript	<code>X<sup>2</sup></code>	X^2
Subscript	<code>H<sub>2</sub>O</code>	H_2O

Note — sub/superscript: CommonMark/GFM has no native sub/superscript syntax. Lattice renders `<sup>` and `<sub>` HTML tags directly (GitHub and Obsidian do too). We recommend, though that you use alternatives that work everywhere: KaTeX math (`$X^2$` $\rightarrow X^2$, `$H_2O$` $\rightarrow H_2O$) or Unicode characters (`x²` , `H₂O`). The linter flags these tags as a portability hint — they may not render in all renderers.

Highlight Marks

Highlight marks are a Lattice-exclusive feature. Wrap any text in `==double equals==` to render it with a **bright amber background**.

This works in **any context** — inside a sentence, `==across italic and bold spans==`, or on its own line:

This entire sentence is highlighted.

[!NOTE] Highlight marks are **not** rendered inside code blocks or inline code — `==this stays raw==` .

Unordered

- First item
- Second item
 - Nested item A
 - Nested item B
 - Deeply nested
- Third item with **bold** and *italic* text

Ordered

1. Install Lattice from the [releases page](#)
2. Open any `.md` file — double-click or via **File** → **Open**
3. Choose your view:
 - i. **Edit** — raw CodeMirror editor only
 - ii. **Preview** — rendered HTML only
 - iii. **Dual** — side-by-side (four layout variants)
4. Print with `Ctrl/Cmd+P` — always renders to a clean light theme

Definition-style (using bold + indent)

Local-first All files stay on your machine. No cloud sync required.

Documents as Code Plain-text `.md` files are diff-friendly, version-controlled, and AI-ready.

Task Lists (GFM)

GitHub Flavored Markdown task lists render as interactive checkboxes in Lattice.

- ☒ Install Lattice
- ☒ Open the demo file
- ☒ Explore Edit / Preview / Dual views
- ☐ Try `Ctrl/Cmd+Shift+T` to regenerate the Table of Contents
- ☐ Try `Ctrl/Cmd+Shift+L` to auto-align a pipe table
- ☐ Paste a screenshot with `Ctrl/Cmd+V` — it auto-saves to a sibling folder
- ☒ Open a second file in a new window (multi-window support)
- ☐ Try printing to PDF with `Ctrl/Cmd+P`

Tables

Lattice supports **GFM pipe tables** with left, center, and right alignment. Press `Ctrl/Cmd+Shift+L` with the cursor anywhere in the table to auto-format it.

Feature	Edit View	Preview View	Dual View
CodeMirror editor	✓	—	✓
Live rendered preview	—	✓	✓
Scroll sync	—	—	✓
Print (Ctrl+P)	✓	✓	✓
Highlight marks	✓	✓	✓
Mermaid diagrams	—	✓	✓
KaTeX mathematics	—	✓	✓

Code

Inline code

Use backticks for `inline code`. Great for `file/paths`, `commands`, and `variable_names`.

Fenced code blocks

Lattice passes code fences to the preview unchanged — syntax highlighting is provided by the renderer.

```
// Lattice highlight-mark plugin – converts ==text== to <mark> elements
export const MARK_RE = /==(?:[^\n]|(?:!=))+?==/g;

export function splitAtMarks(text: string): HastNode[] {
  const parts: HastNode[] = [];
  let lastIndex = 0;
  MARK_RE.lastIndex = 0;
  let match: RegExpExecArray | null;
  while ((match = MARK_RE.exec(text)) !== null) {
    if (match.index > lastIndex)
```

```

    parts.push({ type: 'text', value: text.slice(lastIndex, match.index) });
    parts.push({
      type: 'element', tagName: 'mark',
      properties: {},
      children: [{ type: 'text', value: match[1] }],
    });
    lastIndex = match.index + match[0].length;
  }
  if (lastIndex < text.length)
    parts.push({ type: 'text', value: text.slice(lastIndex) });
  return parts;
}

```

```

// Tauri command – read file with conflict detection
#[tauri::command]
pub async fn read_file(path: String) -> Result<FileReadResult, String> {
  let content = fs::read_to_string(&path)
    .map_err(|e| format!("Cannot read {path}: {e}"))?;
  let hash = compute_sha256(&content);
  Ok(FileReadResult { content, hash })
}

```

```

# Build a production release
export LATTICEBUILD_NO="MyBuild-001"
npm run tauri build

```

Blockquotes

Single-level blockquote. Great for callouts, citations, or side notes.

Nested blockquotes are also supported:

Inner quote — *"Documents as code, not documents as data."*

— Lattice design philosophy

GitHub-style Alerts (rendered by GitHub and Lattice)

[!NOTE] Useful information that users should know, even when skimming.

[!TIP] Optional advice for a better experience.

[!IMPORTANT] Key information users need to know to achieve their goal.

[!WARNING] Urgent information that needs immediate attention.

[!CAUTION] Advises about risks or negative outcomes.

Mathematics — KaTeX

Lattice renders mathematics via **KaTeX**. Use `$...$` for inline and `$$...$$` for display math.

Inline math

Einstein's famous equation $E = mc^2$ and the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ render inline.

The Euler identity $e^{i\pi} + 1 = 0$ is often called the most beautiful equation in mathematics.

Display math — Calculus

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

$$\frac{d}{dx} \left[\int_a^{g(x)} f(t) dt \right] = f(g(x)) \cdot g'(x)$$

Display math — Linear Algebra

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x} \implies \det(\mathbf{A} - \lambda\mathbf{I}) = 0$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Display math — Series & Limits

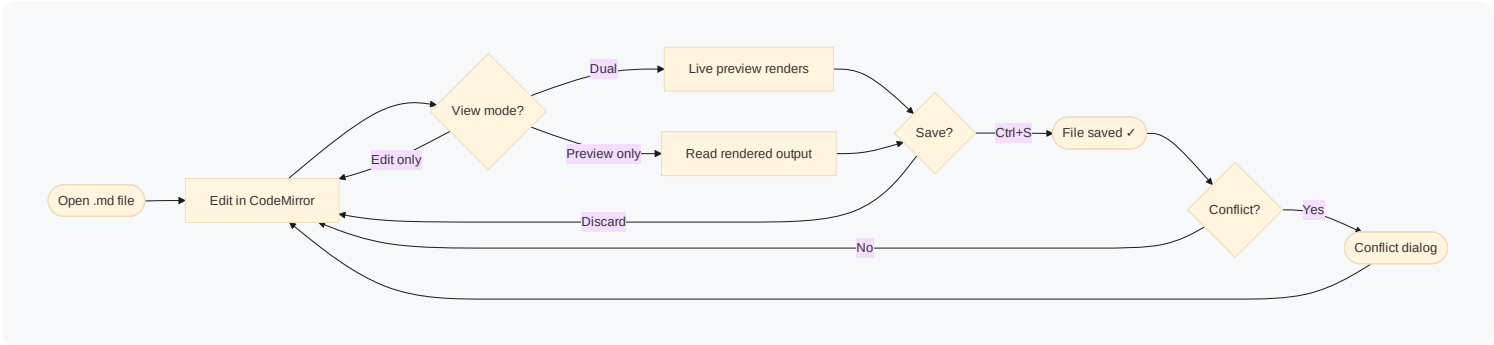
$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e$$

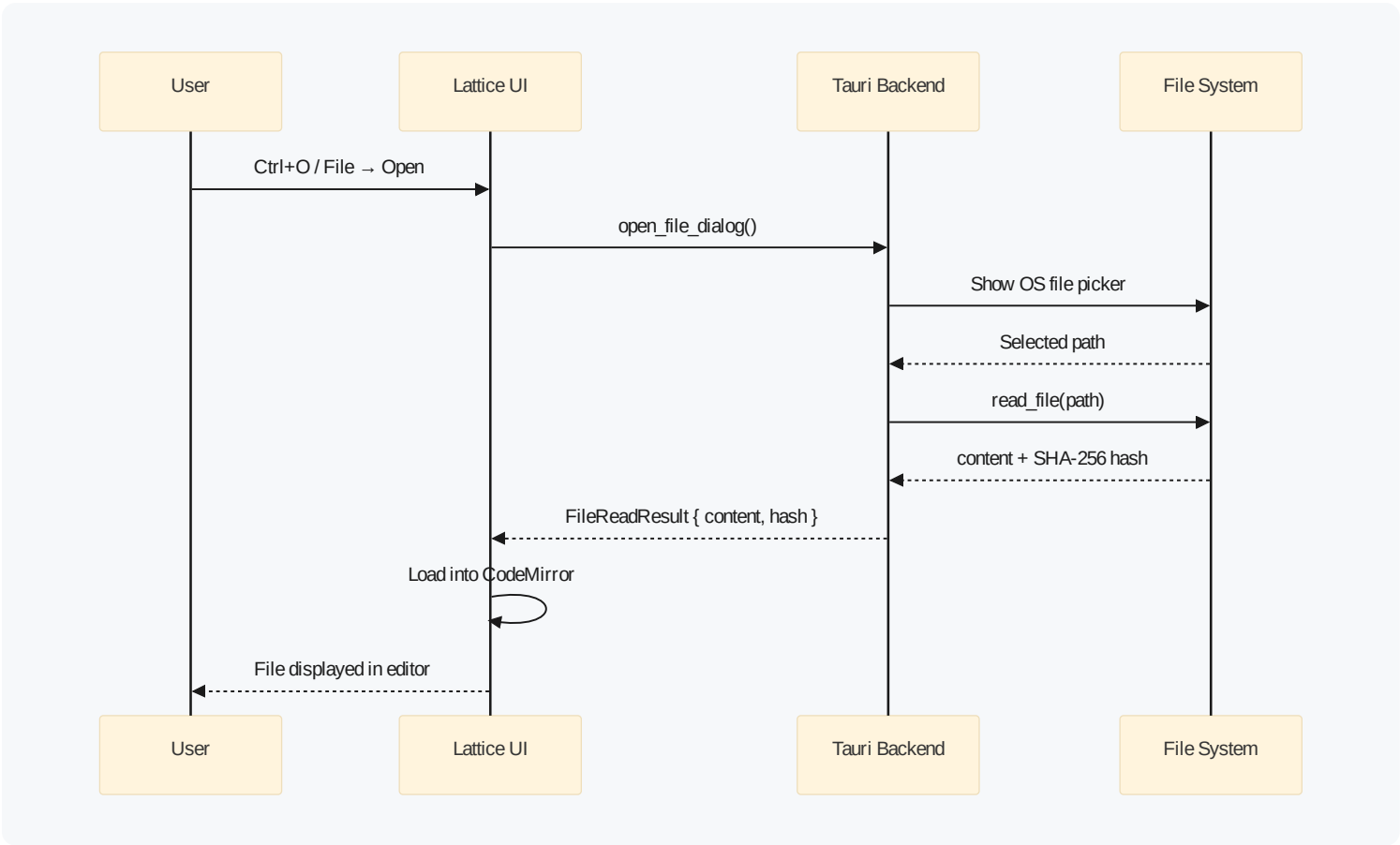
Mermaid Diagrams

Lattice renders ````mermaid```` fenced blocks inline in the preview.

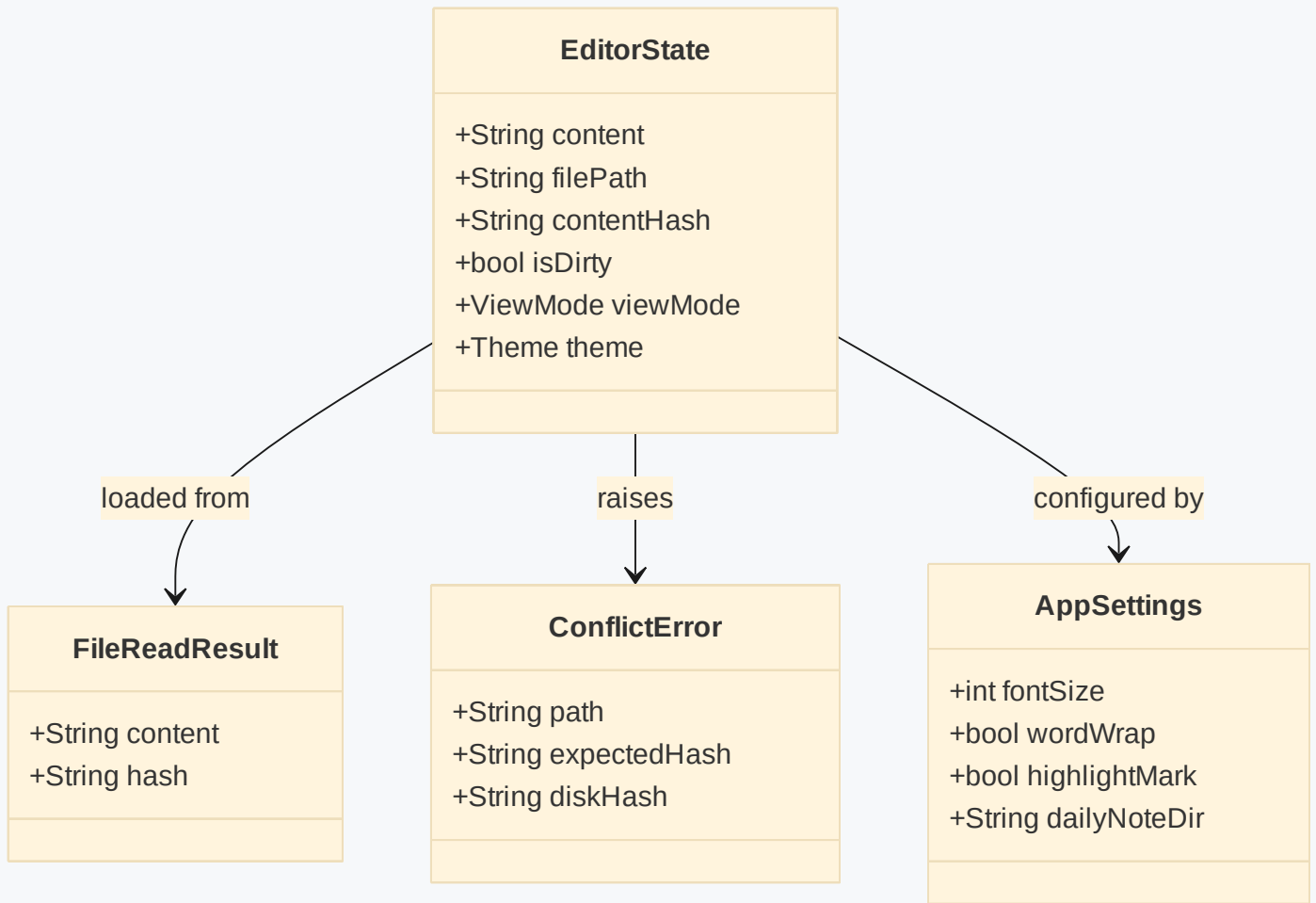
Flowchart — Lattice Edit Cycle



Sequence Diagram — File Open Flow

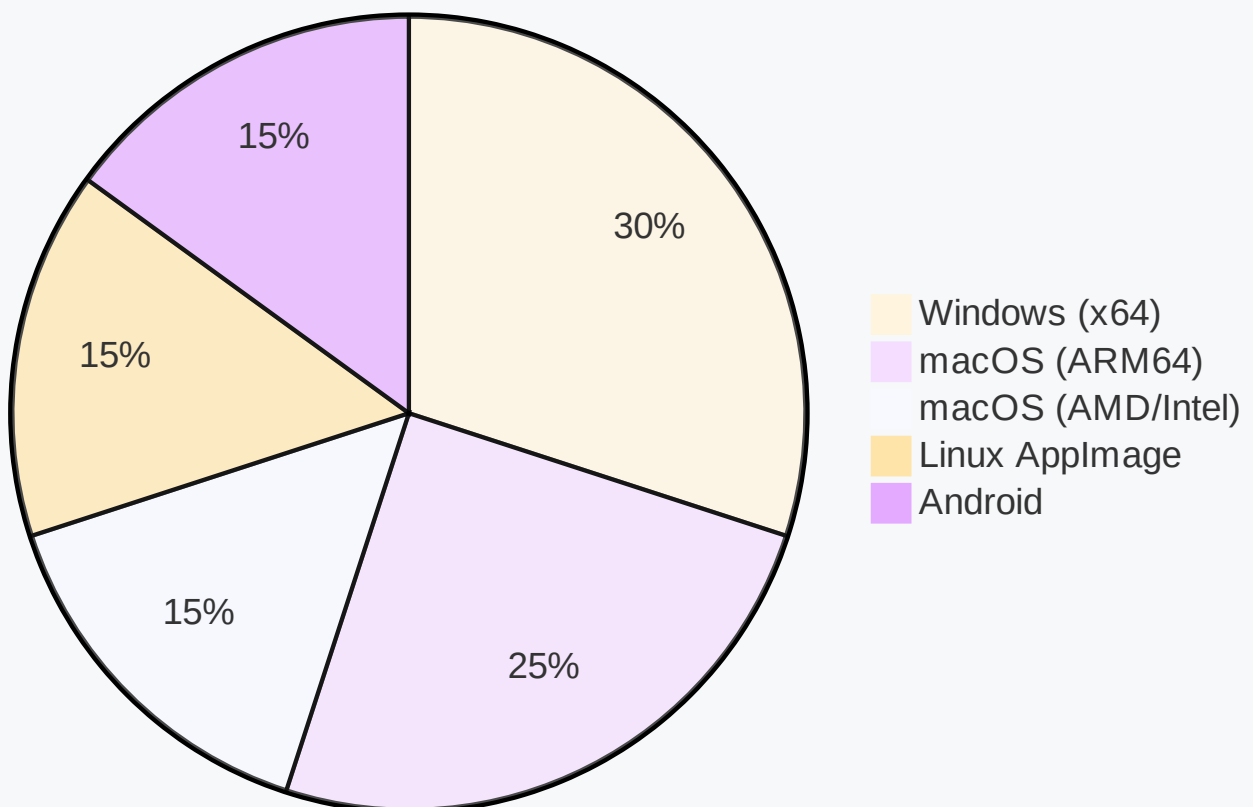


Class Diagram — Core Data Model



Pie Chart — Supported Platforms

Lattice Platform Coverage



Images

Paste images directly into Lattice with `Ctrl/Cmd+V`.

The image file is automatically saved to a sibling folder named after the current document, and a Markdown reference is inserted at the cursor position:

```
! [Image](demo_assets/img_1780574419628.png)
```

! [Image](demo_assets/img_1780575215039.png)

Here is how it renders in `Dual (edit on the left) mode`:

The image shows a dual-pane editor interface. The left pane displays LaTeX source code, and the right pane shows the rendered output.

Left Pane (Source Code):

```
demo.md - lattice (v0.2.39 - windows-desktop-2026W23.0EEC)
%USERPROFILE%\_prj\lattice-ws\lattice\docs\demo\demo.md

207 ## Mathematics — KaTeX
208
209 Lattice renders mathematics via KaTeX. Use $....$ for inline and
    $$...$$ for display math.
210
211 ### Inline math
212
213 Einstein's famous equation  $E = mc^2$  and the quadratic formula
214  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$  render inline.
215
216 The Euler identity  $e^{i\pi} + 1 = 0$  is often called the most
    beautiful equation in mathematics.
217
218 ### Display math — Calculus
219
220 $$
221 \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}
222 $$
223
224 $$
225 \frac{d}{dx} \left[ \int_a^{g(x)} f(t) dt \right] = f(g(x)) \cdot g'(x)
226 $$
227
228 ### Display math — Linear Algebra
229
230 $$
231 \mathbf{A} \mathbf{x} = \lambda \mathbf{x} \implies \det(\mathbf{A} - \lambda \mathbf{I}) = 0
232 \quad \quad \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}
233 \det(\mathbf{A}) - \lambda \det(\mathbf{I}) = 0
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235
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243
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245 e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots
246 = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots
247 $$
248
249 $$
250 \lim_{n \rightarrow \infty} \left( 1 + \frac{1}{n} \right)^n = e
251 $$
252
253 ---
```

Right Pane (Rendered Output):

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$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$
$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e$$

Mermaid Diagrams

Lattice renders `mermaid` fenced blocks inline in the preview.

Flowchart — Lattice Edit Cycle

```
graph TD
    Start([Open and file]) --> Edit[Edit in CodeMirror]
    Edit --> View[View model]
    View --> Render[Render preview]
    Render --> Preview[Preview only]
    Preview --> Edit
    Preview --> Save[Save?]
    Save --> Confirmed[File saved ✓]
    Confirmed --> Edit
    Confirmed --> Conflict{Conflict?}
    Conflict -- No --> Edit
    Conflict -- Yes --> ConflictDialog[Conflict dialog]
```

Here is how it renders in Dual (edit on the bottom) mode:

demo.md - lattice (v0.2.39 · windows-desktop-2026W23.0EEC)

%USERPROFILE%_prj\lattice-ws\lattice\docs\demo\demo.md

-70%+

Dual (edit on bottom)

Menu

Mathematics — KaTeX

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$$\mathbf{Ax} = \lambda \mathbf{x} \implies \det(\mathbf{A} - \lambda \mathbf{I}) = 0$$
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Display math — Series & Limits

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

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223

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225 \frac{d}{dx} \left[\int_a^{g(x)} f(t) \, dt \right] = f(g(x)) \cdot g'(x)

226 \$\$

227

228 ### Display math - Linear Algebra

229

230 \$\$

231 \mathbf{Ax} = \lambda \mathbf{x}

232 \implies \det(\mathbf{A} - \lambda \mathbf{I}) = 0

233

files grouped together and easy to version-control.

Links

- [Lattice on GitHub](#) — source code and issue tracker
- [Releases Page](#) — download installers
- [Release Site](#) — GitHub Pages release portal
- Relative link: [System Architecture Demo](#)
- Relative link: [Physics Equations Demo](#)

Auto-link: <https://github.com/blessia-blessini/lattice>

HTML in Markdown

Lattice renders a **safe whitelist** of inline HTML tags and passes everything else through as literal text. The linter flags all inline HTML — whitelisted tags get a **hint** (portability risk), non-whitelisted tags get a **warning** (not rendered here).

HTML element	Lattice preview	GitHub	Portable alternative
<code><sup></code>	✓ rendered	✓	KaTeX: <code>\$X^2\$</code> → X^2 · Unicode: <code>x²</code>
<code><sub></code>	✓ rendered	✓	KaTeX: <code>\$H_2O\$</code> → H_2O · Unicode: <code>H₂O</code>
<code><kbd></code>	✓ rendered	✓	Inline code: <code>`ctrl+s`</code>
<code>
</code>	✓ rendered	✓	Two trailing spaces or blank line
<code><mark></code>	⚠ plain text	✓	<code>==highlight==</code> (Lattice-native)
<code><details></code>	⚠ plain text	✓	Headings + TOC (no universal equivalent)
Other HTML	⚠ plain text	varies	Markdown equivalent (case-by-case)

Live examples of the whitelisted tags:

- Superscript: $X^2 + Y^3$
- Subscript: H_2O and CO_2

- Keyboard: press `ctrl` + `s` to save

Keyboard Shortcuts Reference

Action	Windows / Linux	macOS
Save	<code>ctrl+s</code>	<code>Cmd+S</code>
Open file	<code>ctrl+o</code>	<code>Cmd+O</code>
New window	<code>ctrl+N</code>	<code>Cmd+N</code>
Generate / refresh TOC	<code>ctrl+Shift+T</code>	<code>Cmd+Shift+T</code>
Auto-align table	<code>ctrl+Shift+L</code>	<code>Cmd+Shift+L</code>
Paste image from clipboard	<code>ctrl+v</code>	<code>Cmd+V</code>
Print / Save as PDF	<code>ctrl+p</code>	<code>Cmd+P</code>
Increase font size	<code>ctrl++</code>	<code>Cmd++</code>
Decrease font size	<code>ctrl+-</code>	<code>Cmd+-</code>